MILITARY SPECIFICATION

RADIO SET AN/GRC-109( )
(Units Of)

This amendment forms a part of Military Specification MIL-R-55242(EL), 21 October 1963

Page 1, paragraph 2.1:

a. Add the following:

"JAN-E-480	Enamel, Baking, Phenol or Urea Formaldehyde

b. Delete the following:

"MIL-I-11748	Interference Reduction for Electrical Equipment"

Page 3, paragraph 3.8 Finish. - Replace second sentence with the following:

"The final paint film on Type 1 surfaces shall be black, semi-gloss enamel in accordance with Specification JAN-E-480 (Type 1), matching a color chip provided by the procuring agency (see 6.3)".

Page 5, paragraph 3.14.2(a) Operating. - Replace first sentence with the following:

"Ambient temperature in the range of +150°F to -40°F for Power Supplies PP-2684( )/GRC-109, PP-2685( )/GRC-109, and Radio Transmitter T-784( )/GRC-109, and ambient temperature in the range of +150°F to +5°F for Radio Receiver R-1004( )/GRC-109. (The +150°F temperature includes effect of sun-load)".

Page 5, paragraph 3.15.2 Sensitivity. - Replace with the following:

"3.15.2 Sensitivity (see 4.8.2) - Radio frequency signal inputs of 1.5 microvolts shall produce a signal plus noise to noise ratio of at least 10 decibels (db) with the receiver adjusted for both AM and CW reception".

Page 6, paragraph 3.16.1 Frequency range. - Replace frequency ranges with the following:

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Band	Frequency
1	3,000 to 5,500 kc
2	5,500 to 9,000 kc
3	9,000 to 16,000 kc
4	16,000 to 22,000 kc

Page 6, paragraph 3.17.1 Power sources. - Replace "6 volt storage battery" with "6.3 to 6.6 volts d.c. from a storage battery or a direct current power supply".

Page 6, paragraph 3.17.3 Ripple voltage. - In third line, insert "(except the ripple voltage shall be less than 10 volts at input voltages of 75 to 260 volts a.c, 40 c.p.s)" after "450 volt output".

Page 7, paragraph 3.17.5 Battery charging output (see 4.10.5) - In the fourth line, delete "at 7 volts".

Page 7, paragraph 3.18.2 Ripple voltage (see 4.11.2) - In third line, insert "(except the ripple voltage shall be less than 18 volts at input voltages of 75 to 260 volts a.c, 40 c.p.s)" after "450 volt output".

Page 7, paragraph 3.22 500 hour operational requirement - Delete in its entirety.

Page 7, paragraph 3.24 Suppression, radio interference. - Delete in its entirety.

Page 8, paragraph 4.3 Preproduction inspection. - In second sentence, delete reference to "group C" and "Table IV".

Page 8, paragraph 4.4 Inspection covered by subsidiary documents. - Delete "Suppression.....3.24".

Page 8, paragraph 4.5 Quality conformance inspection of equipment before preparation for delivery. - In last sentence, delete reference to "group C".

Page 9, paragraph 4.5.3 through 4.5.3.2, and Table IV (Group C Inspection) - Delete in its entirety.

Page 11, paragraph 4.8.2 Sensitivity. -

a. Change first sentence to read:

"The receiver shall be adjusted for standard test conditions except the gain control shall be set to produce an audio output of 2.5 milliwatts or to maximum if this output cannot be obtained".

b. Add at end of paragraph:

"Repeat the above test with the BFO tuned on and the signal generator output 1.5 microvolts unmodulated".

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Page 11, paragraph 4.8.3 I.F. selectivity. - Change second sentence to read:

"With an input signal of 100 microvolts, the gain control shall be set to produce a reference output of 2.5 milliwatts".

Page 11, paragraph 4.8.4 Image rejection. - Change first sentence to read:

"The signal generator output with modulation shall be adjusted to 10 microvolts and the gain control set to produce a reference output of 2.5 milliwatts at each test frequency".

Page 11, paragraph 4.8.5 Spurious response. - Change first sentence to read:

"The signal generator output with modulation shall be adjusted to 10 microvolts and the gain control set to produce a reference output of 2.5 milliwatts at each test frequency".

Page 11, paragraph 4.8.6 I.F. rejection. - Change first sentence to read:

"The signal generator output with modulation shall be adjusted to 10 microvolts and the gain control set to produce a reference output of 2.5 milliwatts at each test frequency".

Page 11, paragraph 4.8.7 Cross signal distortion. - Replace first and second sentences with the following:

"The signal generator output with modulation shall be adjusted to 10 microvolts and the gain control set to produce a reference output of 2.5 milliwatts at each test frequency".

Page 12, paragraph 4.8.12 Plate voltage range. - Change second sentence to read:

"The signal generator output with modulation shall be adjusted to 10 microvolts and the gain control set to produce a reference output of 2.5 milliwatts at each test frequency".

Page 12, paragraph 4.8.14 Audio power output. - Replace with the following:

"4.8.14 Audio power output - The signal generator output with 30 percent modulation of 1000 cps shall be set to 500 microvolts at 3.0 megacycles. The receiver shall be tuned to this signal and the power output measured with the gain control set at maximum output (see 3.15.14)".

Page 12, paragraph 4.8.15 Crystal operation. - Change first sentence to read:

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"The receiver shall be adjusted for standard test conditions except the gain control shall be set to produce an audio output of 2.5 milliwatts or to maximum if this output cannot be obtained".

Page 12, paragraph 4.9.4 Keying speed. - Add at end of paragraph:

"An alternate test method shall be to observe the transmitted signal on an oscilloscope with the Coder-Burst-Transmission Group AN/GRA-71 operating in the Identification (IDY) mode. The rise time and the fall time of the transmitted pulses shall be not more than 10 percent of the pulse width".

Page 13, paragraph 4.12.1 Temperature test. - Replace second sentence and table with the following:

"For Radio Transmitter T-784( )/GRC-109, the power output measurements specified in 4.9.2 shall be made alternately using Power Supply PP-2684( )/GRC-109 and Power Supply PP-2685( )/GRC-109 at Steps 1, 3, 8, and 10. For Radio Receiver R-1004( )/GRC-109, the sensitivity measurements specified in 4.8.2 and the crystal operation measurements specified in 4.8.15 shall be made alternately using Power Supply PP-2684( )/GRC-109 and Power Supply PP-2685( )/GRC-109 at Steps 1, 3, +5°F, and 10".

Page 14, paragraph 4.12.4 Immersion test. - In first sentence, replace the words "covers removed" with "covers installed in place".

Page 15, paragraph 4.13 Air-seal test. - Change first sentence to read:

"All units, with covers removed, shall be opened and closed again in such a manner as to break and remake the seal".

Page 15, paragraph 4.16 500 hour operational test. - Delete in its entirety.

Page 17, paragraph 6.3 Color. - Replace with the following:

"6.3 Color. - The color chip furnished by the procuring agency (3.8) will be color chip No. 27038 and may be obtained upon request to the Commanding General, US Army Electronics Command, ATTN: AMSEL-PP, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103".

Page 17, paragraph 6.6 Group C inspection. - Delete in its entirety.

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